

DIGITAL_TWIN_MODEL.md

SynChain Digital Twin Model

Definition

A Digital Twin is a virtual representation of a real-world supply chain.

Agents never operate directly on real-world data.

Agents operate on the Digital Twin.

Core States

Inventory State

Represents:

- Current Stock
 - Safety Stock
 - Reorder Level
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Demand State

Represents:

- Current Demand
 - Forecast Demand
 - Growth Trends
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Warehouse State

Represents:

- Available Capacity
 - Utilization
 - Location
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Supplier State

Represents:

- Supplier Performance
 - Lead Time
 - Reliability
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Transportation State

Represents:

- Delivery Time
 - Route Information
 - Distribution Efficiency
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Market State

Represents:

- Industry Trends
 - Demand Signals
 - External Factors
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Risk State

Represents:

- Operational Risk
 - Supply Risk
 - Logistics Risk
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Twin Lifecycle

1. User submits data
2. Digital Twin created
3. Agents analyze twin

4. Twin updated
 5. Recommendations generated
 6. User reviews results
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Future Twin Enhancements

- Real-time updates
- Live supplier feeds
- IoT integration
- ERP integration
- Predictive procurement